

['LASTracker: A lightweight RGB-E tracking framework with ANN-SNN adaptive switching',

<https://www.sciencedirect.com/science/article/pii/S0031320326005881>,

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'Zikai Wang, Siying Liu, Hanle Zheng, Sihan Wang, Yifan Hu, Hanxiao Fan, Yao Li, Hao Guo, Lei Deng',

'2',

'2026',

'11'],

['Causality-Based Modality- and Platform-Invariant Representation Learning for Dynamic RGBT Tracking and a Benchmark',

<http://ieeexplore.ieee.org/document/11449406>,

'Each sequence in existing RGBT tracking datasets is typically captured from a single platform equipped with both RGB (visible light) and TIR (thermal infrared) sensors. In real-world applications, tracking some objects requires cross-platform collaboration and these platforms might be equipped with different sensors. However, changes in modalities and platforms may cause significant variations in target appearance and abrupt position shifts, which existing RGBT trackers struggle to handle. To address these challenges, we define a new task, termed dynamic RGBT tracking, focusing on cross-platform and modality-variant scenarios. Considering the dynamic changes of modalities and platforms, we investigate dynamic RGBT tracking from a causal perspective, and assume that images consist of causal factors (target-relevant information) and non-causal factors (target-irrelevant information, i.e., modality/platform information), where only the former is conducive to stable tracking. Based on this assumption, we propose a novel causality-based modality&platform-invariant representation learning approach to capture robust invariant representations for dynamic RGBT tracking. In particular, to mitigate the challenges posed by modality variations, we design a causal consistency encoder that introduces an intervener to model feature uncertainty and simulate modal variations, compelling the model to focus on modality-invariant features to improve tracking robustness. To overcome the issue of abrupt view change and position shift, we design a platform-independent global searcher to re-localize the target whenever a platform switch occurs, which leverages an intervener to simulate the interference of platform changes on features, encouraging the searcher to learn platform-invariant representations for improved localization accuracy. In addition, to promote the research and development of dynamic RGBT tracking, we construct a dataset named DRGBT603, which consists of 603 sequences with a total of 1.49 M frame pairs. Extensive experiments on DRGBT603 dataset validate the effectiveness of the proposed method against other state-of-the-art methods. Our code and data are now available:

<https://github.com/dongdong2061/DRGBT>,

'Zhaodong Ding, Chenglong Li, Shengqing Miao, Jin Tang',

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'4'],

['Spatial-Temporal Scene Graph Generation for Open-Vocabulary Multiple Object Tracking',

<http://ieeexplore.ieee.org/document/11457845>,

'Zhiqiang Zhao, Daitu Wen, Yuanhang Gu, Xiaoli Luo, Tao Ma, Xu Ma, Bin Wu',

'6',

'2026',

'3']]

['HMamba-3DFT: A hierarchical mamba framework for emotion-driven semantic 3D facial tracking',

<https://www.sciencedirect.com/science/article/pii/S0031320326003808>,

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'Haodong Jin, Muwei Jian, Derui Ding, Hui Yu',

'2',

'2026',

'10'],

['Multi-modal object tracking with detailed spatial prompt learning',

<https://www.sciencedirect.com/science/article/pii/S0031320326003985>,

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'Gang He, Long Gao, Yuze Ke, Haijun Chen, Yan Jiang',

'2',

'2026',

'10'],

['RingMoE: Mixture-of-Modality-Experts Multi-Modal Foundation Models for Universal Remote Sensing Image Interpretation',

<http://ieeexplore.ieee.org/document/11298536>,

'The rapid advancement of foundation models has revolutionized visual representation learning in a self-supervised manner. However, their application in remote sensing (RS) remains constrained by a fundamental gap: existing models predominantly handle single or limited modalities, overlooking the inherently multi-modal nature of RS observations. Optical, synthetic aperture radar (SAR), and multi-spectral data offer complementary insights that significantly reduce the inherent ambiguity and uncertainty in single-source analysis. To bridge this gap, we introduce RingMoE, a unified multi-modal RS foundation model with 14.7 billion parameters, pre-trained on 400 million multi-modal RS images from nine satellites. RingMoE incorporates three key innovations: 1) A hierarchical Mixture-of-Experts (MoE) architecture comprising modal-specialized, collaborative, and shared experts, effectively modeling intra-modal knowledge while capturing cross-modal dependencies to mitigate conflicts between modal representations; 2) Physics-informed self-supervised learning, explicitly embedding sensor-specific radiometric characteristics into the pre-training objectives; 3) Dynamic expert pruning, enabling adaptive model compression from 14.7B to 1B parameters while maintaining performance, facilitating efficient deployment in Earth observation applications. Evaluated across 23 benchmarks spanning six key RS tasks (i.e., classification, detection, segmentation, tracking, change detection, and depth estimation), RingMoE outperforms existing foundation models and sets new SOTAs, demonstrating remarkable adaptability from single-modal to multi-modal scenarios. Beyond

theoretical progress, it has been deployed and trialed in multiple sectors, including emergency response, land management, marine sciences, and urban planning.'

'Hanbo Bi, Yingchao Feng, Boyuan Tong, Mengyu Wang, Haichen Yu, Yongqiang Mao, Hao Chang, Wenhui Diao, Peijin Wang, Yue Yu, Hanyang Peng, Yehong Zhang, Kun Fu, Xian Sun',

'3',

'2026',

'3'],

['BEVTrack: Multi-View Multi-Human Registration and Tracking in the Bird's Eye View',

<http://ieeexplore.ieee.org/document/11314795>'],

'We handle a new problem of multi-view multi-human tracking in the bird's eye view (BEV). Different from previous works, we require neither the calibration among the multi-view cameras nor the actually captured BEV video. This makes the studied problem closer to real-world applications, however, more challenging. For this purpose, in this work, we propose a novel BEVTrack scheme. Specifically, given multi-view videos, we first use a virtual BEV transform module to obtain the BEV for each view. Then, we propose a unified BEV alignment module to fuse the respectively generated BEVs, in which we specifically design the self-supervised losses by considering both the spatial consistency and the temporal continuity. During the inference, we design the camera-subject collaborative registration and tracking strategy to make use of the mutual dependence between the multi-view cameras and the multiple targets, to achieve the desired BEV tracking. We also build a new benchmark for training and evaluation, the experimental results on which have verified the rationality of the problem and the effectiveness of our method.'

'Zekun Qian, Wei Feng, Feifan Wang, Ruize Han',

'3',

'2026',

'3'],

['Causal HyperPrompter: A Framework for Unbiased Hyperspectral Camouflaged Object Tracking',

<http://ieeexplore.ieee.org/document/11314773>'],

'Hyperspectral camouflaged object tracking remains a significant challenge due to the high similarity between objects and replicas in texture and color. Despite recent progress, the bias present in the tracker and the embedding token hinders the model training. Specifically, most methods rely on false-color three-channel images to fine-tune RGB-based trackers. However, it introduces a confounding effect within the RGB domain, potentially leading to harmful biases that misguide the model toward spurious correlations while neglecting the critical spectral discrimination inherent in hyperspectral images. Furthermore, current token-type embedding methods overlook the key correlations between templates and searches, ultimately confusing correlation and impairing tracking performance. To address these challenges, this paper proposes a new unbiased tracking framework named Causal HyperPrompter. It first introduces a structural causal model to disentangle and control exclusive causal factors during tracking, and incorporates a counterfactual intervention strategy to eliminate confounding variables and mitigate the bias inherited from RGB-based models. In addition, we present a novel token-type embedding module that integrates local spectral angle modeling to enhance the semantic link between template and search tokens, thereby improving the model's

sensitivity to object localization. Lastly, to overcome the difficulty of manually initializing the bounding box and addressing data scarcity, we introduce a large-scale hyperspectral camouflaged object detection and tracking dataset, BihoT-130 k, consisting of 1,30,750 annotated frames across various camouflage scenes. Extensive experiments on multiple large-scale datasets illustrate the effectiveness of our proposed methods.',

'Hanzheng Wang, Wei Li, Xiang-Gen Xia, Qian Du',

'3',

'2026',

'3'],

['ASDTracker: Adaptively Sparse Detection With Attention-Guided Refinement for Efficient Multi-Object Tracking',

<http://ieeexplore.ieee.org/document/11396433>'],

'Tracking-by-Detection paradigms shine in generic multi-object tracking (MOT), while their compact construction hinders the real-time applications. In this work, we attribute the substantial computational burden to two expensive components, i.e. detection and re-identification. Building upon the principle of adaptively maintaining acceptable inference efficiency, we present Adaptively Sparse Detection with attention-guided refinement (ASDTracker) for efficient tracking. In specific, our ASDTracker rapidly assess the short-term and long-term occlusion, dynamically determining the usage of the expensive detector. For non-key frames, we efficiently refine small-size crops out of Kalman Filter predictions and introduce the noisy shadow labels to robustly train this refinement network. Additionally, we substitute the lightweight appearance representation for the heavy ReID network, which efficiently extracts sufficient appearance cues in the coarsely quantized color spaces. Extensive experiments on four benchmarks demonstrate that ASDTracker achieves competitive performance in generalization and robustness under favorable inference speed. Moreover, the efficient tracking deployment is further implemented to an unmanned surface vehicle with high accuracy and low latency in real-world scenarios.',

'Yueying Wang, Chenyang Yan, Cairong Zhao, Weidong Zhang, Dan Zeng',

'4',

'2026',

'3'],

['Monocular Multi-Object 3D Visual Language Tracking',

<http://ieeexplore.ieee.org/document/11391514>'],

'Visual Language Tracking (VLT) enables machines to perform tracking in real world through human-like language descriptions. However, existing VLT methods are limited to 2D spatial tracking or single-object 3D tracking and do not support multi-object 3D tracking within monocular video. This limitation arises because advancements in 3D multi-object tracking have predominantly relied on sensor-based data (e.g., point clouds, depth sensors) that lacks corresponding language descriptions. Moreover, natural language descriptions in existing VLT literature often suffer from redundancy, impeding the efficient and precise localization of multiple objects. We present the first technique to extend VLT to multi-object 3D tracking using monocular video. We introduce a comprehensive framework that includes (i) a Monocular Multi-object 3D Visual Language Tracking (MoMo-3DVL

task, (ii) a large-scale dataset, MoMo-3DRoVLT, tailored for this task, and (iii) a custom neural model. Our dataset, generated with the aid of Large Language Models (LLMs) and manual verification, contains 8,216 video sequences annotated with both 2D and 3D bounding boxes, with each sequence accompanied by three freely generated, human-level textual descriptions. We propose MoMo-3DVLTracker, the first neural model specifically designed for MoMo-3DVL. This model integrates a multimodal feature extractor, a visual language encoder-decoder, and modules for detection and tracking, setting a strong baseline for MoMo-3DVL. Beyond existing paradigms, it introduces a task-specific structural coupling that integrates a differentiable linked-memory mechanism with depth-guided and language-conditioned reasoning for robust monocular 3D multi-object tracking. Experimental results demonstrate that our approach outperforms existing methods on the MoMo-3DRoVLT dataset. Our dataset and code are available at <https://github.com/hongkai-wei/MoMo-3DVL>.'

'Hongkai Wei, Rong Wang, Haixiang Hu, Shijie Sun, Xiangyu Song, Mingtao Feng, Keyu Guo, Yongle Huang, Hua Cui, Naveed Akhtar',

'4',

'2026',

'3'],

['SMTrack: State-Aware Mamba for Efficient Temporal Modeling in Visual Tracking',

<http://ieeexplore.ieee.org/document/11408128>'],

'Visual tracking aims to automatically estimate the state of a target object in a video sequence, which is challenging especially in dynamic scenarios. Thus, numerous methods are proposed to introduce temporal cues to enhance tracking robustness. However, conventional CNN and Transformer architectures exhibit inherent limitations in modeling long-range temporal dependencies in visual tracking, often necessitating either complex customized modules or substantial computational costs to integrate temporal cues. Inspired by the success of the state space model, we propose a novel temporal modeling paradigm for visual tracking, termed State-aware Mamba Tracker (SMTrack), providing a neat pipeline for training and tracking without needing customized modules or substantial computational costs to build long-range temporal dependencies. It enjoys several merits. First, we propose a novel selective state-aware space model with state-wise parameters to capture more diverse temporal cues for robust tracking. Second, SMTrack facilitates long-range temporal interactions with linear computational complexity during training. Third, SMTrack enables each frame to interact with previously tracked frames via hidden state propagation and updating, which releases computational costs of handling temporal cues during tracking. Extensive experimental results demonstrate that SMTrack achieves promising performance with low computational costs.'

'Yinchao Ma, Dengqing Yang, Zhangyu He, Wenfei Yang, Tianzhu Zhang',

'4',

'2026',

'3'],

['A Multi-Level Self-Distillation-Based Unified Tracker for Efficient RGB-T Tracking',

<http://ieeexplore.ieee.org/document/11415354>'],

'RGB-Thermal (RGB-T) tracking enhances visual tracking robustness by combining RGB and thermal infrared (TIR) modalities, addressing limitations of RGB-only trackers under challenging conditions such as low light and appearance variations. However, most existing RGB-T trackers rely on complex fusion modules or modality-specific architectures, sacrificing efficiency for performance. In this paper, we propose a novel Multi-level Self-Distillation (MSD) framework that adapts a one-stream RGB tracker to the RGB-T setting without modifying the network architecture or adding any extra parameters. RGB and TIR inputs are jointly processed through a shared backbone, and training is guided by a combination of self-supervised and supervised objectives to enhance cross-modal feature representation. The self-supervised component includes a contrastive loss that aligns semantically consistent regions across template-search pairs, as well as a modality-gap alignment loss that reduces discrepancies between RGB and TIR features. These internal signals complement task-driven supervision, including an intermediate focal loss that strengthens early localization by enhancing shallow and mid-level features, modality-specific losses that preserve distinctive cues under partial modality degradation, and a fused tracking loss that drives final bounding box prediction. Comprehensive evaluations on LasHeR, RGBT234, and GTOT benchmarks demonstrate that MSD achieves state-of-the-art tracking accuracy while maintaining the computational efficiency of the original RGB tracker. Our work establishes a new paradigm in multi-modal tracking by demonstrating that optimized training strategies can outperform complex architectural modifications, offering significant practical advantages for real-world deployment.'

'Mohamed Awad, Ahmed Elliethy, M. Omair Ahmad, M. N. S. Swamy',

'4',

'2026',

'3'],

['SAFT: Real-Time Tracking and Mapping With Self-Supervised Robust Stereo Matching for Underwater Vehicles',

<http://ieeexplore.ieee.org/document/11239517>'],

'Robust and efficient tracking and mapping are critical for underwater vehicles, but remain challenging due to degraded visual quality, ambiguous features, and limited computational resources. Although recent deep learning-based stereo matching methods have significantly improved geometric perception for robots, most existing approaches struggle to simultaneously achieve high speed and strong generalization. To address these challenges, we propose SAFT, a tracking and mapping framework based on self-supervised, robust, and real-time stereo matching. SAFT introduces three key innovations: 1) SAFT-Stereo, a novel stereo matching network that integrates cost aggregation with iterative optimization to enable efficient disparity estimation in feature-sparse regions; 2) a spatiotemporal self-supervised loss that leverages both spatial and temporal constraints to provide stable training signals in textureless regions; and 3) SAFT-DSOL, a real-time tracking and mapping algorithm that integrates the self-supervised models to achieve robust localization and dense reconstruction. Extensive experiments on both public and custom underwater datasets demonstrate that SAFT-Stereo achieves the best generalization performance among all real-time methods, while requiring only 1/6 of the inference time of RT-IGEV++. Moreover, the proposed SAFT-DSOL enables stable and efficient tracking and achieves real-time dense reconstruction in indoor shipwreck scenarios. The code is available at github.com/c237814486/SAFT-Stereo,'

'Yaozhong Cao, Xiaolong Hui, Xuejian Bai, Yu Wang, Shuo Wang, Min Tan',

'5',

'3'],

['MMCATrack: Multi-Modal Channel Attention Tracker',

<https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/cvi2.70060?af=R>,

'We propose a multi-modal channel attention tracking algorithm wherein a multi-modal channel attention block is incorporated for the purpose of enhancing the representation ability of the key features within the multi-modal features. \n\n\n\n\n\n\nABSTRACT\nMost existing Transformer-based visual object tracking methods rely exclusively on the feature map from the last encoder layer for object prediction, thereby overlooking the rich information contained in shallow and intermediate layer feature maps. This limitation reduces the representational capacity of the model. Moreover, current multi-modal tracking frameworks typically construct multi-modal features through simple concatenation, which fails to adequately account for the differential contributions of individual modalities to the final prediction task. As a result, these approaches exhibit an insufficient ability to express key features within the multi-modal representation. To address the aforementioned issues, this paper proposes a multi-modal channel attention tracking algorithm, where a multi-modal channel attention block is incorporated for the purpose of enhancing the representation ability of the key features within the multi-modal features. Specifically, the multi-modal channel attention block first aggregates multi-modal information from the multi-layer feature maps of the encoder through cross layer cascading and then applies channel attention mechanism to dynamically calibrate the channel weights in the generated multi-modal features, thereby enhancing the representation of key features. In addition, this article proposes a new regression loss function to improve localisation accuracy. Finally, abundant experiments conducted on five benchmarks including GOT-10K, TrackingNet, TNL2K, VisEvent and RGBT234 have verified the effectiveness of our theory.'

'Zhiqiang Zhao, Daitu Wen, Yuanhang Gu, Xiaoli Luo, Tao Ma, Xu Ma, Bin Wu',

'6',

'2026',

'3'],

['Advancing Traffic Safety in Dense Urban Settings: A Robust Real-Time Wrong-Way Driving Detection System with Hough Transform and YOLOv5',

<https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ipr2.70322?af=R>,

'The study presents a real-time wrong-way driving detection system that integrates Hough Line Transform for road boundary extraction and YOLOv5 for vehicle detection, achieving 96% accuracy and a 0.05% false positive rate. Validated in Chattogram, Bangladesh, the system demonstrates a 60.05% detection rate across diverse urban traffic scenarios using a novel dataset of 100 videos. This innovative approach enhances traffic safety in dense urban settings by addressing the scarcity of context-specific data.'

ABSTRACT

In Bangladesh, where a burgeoning population strains limited road networks, wrong-way driving (WWD) exacerbates traffic congestion and elevates accident risks, posing a critical public safety challenge. We present a pioneering real-time WWD detection system that integrates cutting-edge image processing and deep learning to transform traffic monitoring. Our methodology combines the Hough line transform for precise road boundary extraction with the state-of-the-art YOLOv5 algorithm for high-fidelity, real-time vehicle detection. Vehicle direction is inferred from single-frame orientation (distinguishing front/rear views) without multi-frame tracking, ensuring lightweight operation and scalable deployment. Tested on a novel,

'2'],

['A Novel Quantised Image Sensing for Machine Vision',

<https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ell2.70515?af=R>,

This paper shows that the image information quantised in the range of 1 to 4 bits can be utilised for object detection and tracking without any performance degradation. Thus, it is expected that the proposed technique holds promise for use in machine vision, not only maintaining the accuracy of detection and tracking, but even enabling more optimised information transfer between the image sensor and the computing system.

ABSTRACT

This paper proposes a quantisation-driven data-compression framework for machine vision sensors by combining nonlinear quantisation and bit-shift masking. The proposed approach compresses pixel data to an effective 1–4 bits per pixel while preserving edge-salient information required for object detection and feature tracking. Experimental results show that the compressed/quantised images maintain detection and tracking performance without degradation, indicating the proposed framework can reduce sensor-to-processor bandwidth and optimise required bit resolution.'

'Paul K. J. Park',

'8',

'2026',

'1']]

august_articles = None

with open("./collected_articles/2026-04.dat", 'rb') as file:

```
august_articles = pickle.load(file)
```

```
search_word = "tracking"
```

```
[article for article in august_articles if (search_word in article[0].lower()) or (search_word in article[2].lower())]
```

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['Causality-Based Modality- and Platform-Invariant Representation Learning for Dynamic RGBT Tracking and a Benchmark',

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'Each sequence in existing RGBT tracking datasets is typically captured from a single platform equipped with both RGB (visible light) and TIR (thermal infrared) sensors. In real-world applications, tracking some objects requires cross-platform collaboration and these platforms might be equipped with different sensors. However, changes in modalities and platforms may cause significant variations in target appearance and abrupt position shifts, which existing RGBT trackers struggle to handle. To address these challenges, we define a new task, termed dynamic RGBT tracking, focusing on cross-platform and modality-variant scenarios. Considering the dynamic changes of modalities and platforms, we investigate dynamic RGBT tracking from a causal perspective, and assume that images consist of causal factors (target-relevant information) and non-causal factors (target-irrelevant information, i.e., modality/platform information), where only the former is conducive to stable tracking. Based on this assumption, we propose a novel causality-based modality&platform-invariant representation learning approach to capture robust invariant representations for dynamic RGBT tracking. In particular, to mitigate the challenges posed by modality variations, we design a causal consistency encoder that introduces an intervener to model feature uncertainty and simulate modal variations, compelling the model to focus on modality-invariant features to improve tracking robustness. To overcome the issue of abrupt view change and position shift, we design a platform-independent global searcher to re-localize the target whenever a platform switch occurs, which leverages an intervener to simulate the interference of platform changes on features, encouraging the searcher to learn platform-invariant representations for improved localization accuracy. In addition, to promote the research and development of dynamic RGBT tracking, we construct a dataset named DRGBT603, which consists of 603 sequences with a total of 1.49 M frame pairs. Extensive experiments on DRGBT603 dataset validate the effectiveness of the proposed method against other state-of-the-art methods. Our code and data are now available:

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'2026',

'4'],

['Spatial-Temporal Scene Graph Generation for Open-Vocabulary Multiple Object Tracking',

<http://ieeexplore.ieee.org/document/11457845>'],

'Open-vocabulary multiple object tracking (MOT) aims to track arbitrary objects in the real world. Although significant progress has been achieved in object classification by leveraging the knowledge from large vision-language models, advances in data association for open-vocabulary MOT remain limited. Existing methods primarily rely on appearance cues to establish associations. However, these cues are often unreliable in the face of occlusions and ambiguous object appearances, resulting in suboptimal tracking performance in complex scenarios. In this paper, we propose a novel open-vocabulary MOT method, Spatial-temporal Scene Graph Tracker (SSGTrack), which introduces a fundamentally different approach to data association by building a Spatial-temporal Scene Graph (SSG) that captures rich semantic and spatial relationships between objects across adjacent frames. Specifically, SSGTrack constructs proposal-level relationships by extracting diverse contextual information from the multi-head self-attention layers of the Transformer decoder. These

<https://ietresearch.onlinelibrary.wiley.com/doi/10.1049/ell2.70551?af=R>,

'Dominant acoustic features are tracked to form compact signal portraits representing oscillatory modes. Faults are detected by portrait deviation, enabling low-complexity condition monitoring of moving machinery.'

ABSTRACT A resource-efficient semi-analytical method for mechanical fault detection via acoustic signal analysis is presented. A compact signal portrait is constructed by tracking dominant salient-frequency peaks over time, without a priori assumptions on peak locations. Peak feature statistics are aggregated into a 3D matrix that encodes nominal oscillatory modes. Fault detection is achieved by measuring the deviation between current and healthy portraits using a single decision threshold. Experiments on internal combustion engine recordings demonstrate improved separability and lower computational cost compared to empirical mode decomposition, confirming suitability for resource-constrained monitoring.'

'Oleksandr Volkov, Oleksandr Bushma, Artem Sieriebriakov, Volodymyr Taranukha, Yaroslav Linder',

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'Paul K. J. Park',

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'Each sequence in existing RGBT tracking datasets is typically captured from a single platform equipped with both RGB (visible light) and TIR (thermal infrared) sensors. In real-world applications, tracking some objects requires cross-platform collaboration and these platforms might be equipped with different sensors. However, changes in modalities and platforms may cause significant variations in target appearance and abrupt position shifts, which existing RGBT trackers struggle to handle. To address these challenges, we define a new task, termed dynamic RGBT tracking, focusing on cross-platform and modality-variant scenarios. Considering the dynamic changes of modalities and platforms, we investigate dynamic RGBT tracking from a causal perspective, and assume that images consist of causal factors (target-relevant information) and non-causal factors (target-irrelevant information, i.e., modality/platform information), where only the former is conducive to stable tracking. Based on this assumption, we propose a novel causality-based modality&platform-invariant representation learning approach to capture robust invariant representations for dynamic RGBT tracking. In particular, to mitigate the challenges posed by modality variations, we design a causal consistency encoder that introduces an intervener to model feature uncertainty and simulate modal variations, compelling the model to focus on modality-invariant features to improve tracking robustness. To overcome the issue of abrupt view change and position shift, we design a platform-independent global searcher to re-localize the target whenever a platform switch occurs, which leverages an intervener to simulate the interference of platform changes on features, encouraging the searcher to learn platform-invariant representations for improved localization accuracy. In addition, to promote the research and development of dynamic RGBT tracking, we construct a dataset named DRGBT603, which consists of 603 sequences with a total of 1.49 M frame pairs. Extensive experiments on DRGBT603 dataset validate the effectiveness of the proposed method against other state-of-the-art methods. Our code and data are now available:

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'Zhaodong Ding, Chenglong Li, Shengqing Miao, Jin Tang',

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'2026',

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